

Population flow reconstruction of an acute human wound interval: held-out evaluation, time coordinates and donor-count sensitivity

Zeyu Fu

Abstract

A smooth trajectory fitted to cross-sectional single-cell data does not establish prediction of an individual’s wound course. We tested whether population fields improve on no motion when a target time point or donor is excluded from fitting. Three healthy donors contributed intact skin and acute wounds at days 1, 7 and 30. A frozen expression representation, training-only standardization and donor-paired endpoint sampling supported separate timepoint-holdout and donor-transfer protocols. We compared four time coordinates, shared and donor-conditioned flow matching, three non-neural predictors, two training-donor counts and three training seeds. Three mouse model arms supplied a separate computational audit. Pooled Wound7 reconstruction reduced energy distance relative to stay-still by +49.9%. Excluding each donor preserved the result for the retained donors in 3/3 sensitivity runs; genuine held-out-donor transfer separately improved by a mean +75.9% in 3/3 donors. Linear days failed at the target (−18.2%), whereas rank, square-root and log-day coordinates improved. All five predictive methods met the specified split-half-reference criterion in 3/3 donor folds. Shared flow had the smallest mean error, 0.395; conditioning improved all six training-donor comparisons without improving the overall held-out mean. Adding a second training donor reduced seed-0 mean error from 0.5496 to 0.3854. Aggregate reductions were 0.156–0.176 across seeds, with donor-block interval [0.081, 0.315], but some individual configurations varied by 0.368. Mouse-arm changes ranged from −1.5% to +50.6%, with one animal per arm and time point. These experiments support reconstruction of one acute human interval in a three-donor pilot, conditional on the representation and evaluation protocol. They establish neither DFU prognosis nor a population-level method ranking, identifiable biological trajectories, or a required cohort size.

Keywords: conditional flow matching; wound healing; held-out donor; energy distance; time coordinate; donor heterogeneity.

1 Introduction

Wound repair involves changing mixtures of inflammatory, stromal and epithelial states over time [17]. In diabetic foot disease, clinical healing and recurrence are outcomes of interest, but they cannot be inferred directly from a transcriptional trajectory [1]. Destructive single-cell sampling measures distributions at selected times rather than following the same cells longitudinally. Trajectory inference can organize these states, yet its conclusions depend on sampling design, topology and modeling assumptions [15, 22].

Optimal transport and continuous-flow models offer ways to connect observed distributions [9, 16, 20, 21]. Neural ordinary differential equations parameterize a continuous field, and flow matching learns such a field from chosen probability paths [4, 10]. Related transport approaches also predict cellular responses to perturbations [3]. Reconstructing a target distribution and identifying an individual biological trajectory remain distinct objectives because multiple dynamics can produce the same marginals [8, 22].

A second distinction concerns generalization. Predicting a time point missing from every donor removes different information from predicting an unseen donor whose target time is already represented in training. Both tasks depend on the expression coordinates and preprocessing; transferable latent representations are useful, but do not eliminate representation choice or sampling effects [7, 12]. Biological replication also remains at the donor level even when many cells enter a distributional comparison [23].

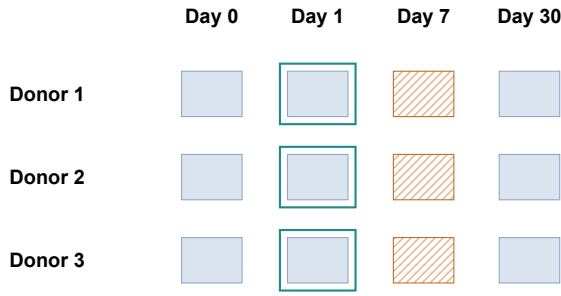
The public acute human wound atlas offers a bounded setting in which to separate these questions: three healthy volunteers contribute skin and wound samples at four times [11, 13]. We ask whether a predicted distribution approaches a held-out target more closely than an unchanged source distribution. We compare missing-time and missing-donor protocols, time coordinates, neural and non-neural predictors, training-donor counts and training seeds. Numerical-resolution and evaluation-sampling analyses test additional sources of computational sensitivity, while three mouse arms supply a separate model comparison. These experiments assess distribution reconstruction and its limits; the available cohorts do not contain a diabetic-foot-ulcer time-to-closure target.

2 Results

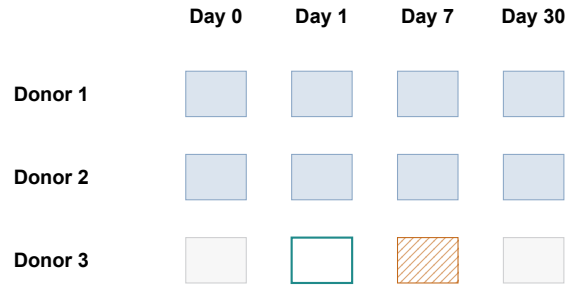
2.1 A held-out acute human time point can be reconstructed under the fixed protocol

The evaluation architecture is summarized in Figure 1. The protocols separate a globally missing Wound7 time point from transfer to a donor excluded from fitting; both use training-only scaling and compare the predicted distribution with stay-still.

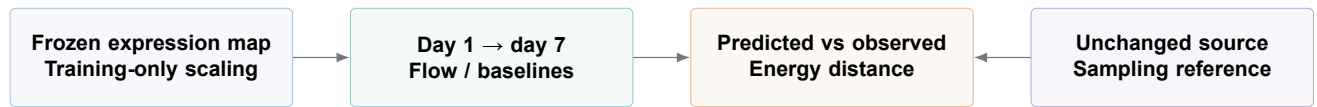
The finalized held-out Wound7 analysis improved on the stay-still baseline by +49.9%. Dropping one donor at a time preserved improvement in 3/3 retained-donor sensitivity runs; these runs predict the retained donors, not the omitted donor. The held-out donor transfer analysis also passed in 3/3 folds, with a mean improvement of +75.9% and a mean pairwise displacement cosine

A Missing time point

Every day-7 cell is excluded from training.
All donors contribute the other time points.

B New donor: example fold

All cells of the test donor are excluded from training.
Its observed day-1 cells initialize prediction.

**C Common evaluation and complementary sensitivity analyses**

Time coordinates · training donors · training seeds · numerical resolution · evaluation sampling

Three healthy human donors; separate mouse comparison; no DFU healing endpoint

Figure 1. Information available in the two temporal prediction tasks. Three healthy human donors contribute skin and wound specimens on days 0, 1, 7 and 30. (A) Global timepoint holdout excludes every day-7 cell from model fitting and scaling. Blue tiles identify training samples; teal outlines also identify the observed day-1 source of prediction; hatched orange tiles are evaluation targets. (B) One illustrative donor-transfer fold excludes all cells of the third donor from fitting and scaling. Only its day-1 cells initialize prediction, its day-7 cells are evaluated, and its grey day-0/day-30 samples are unused. All four times remain available in the other two donors, and the test-donor role is rotated across all three. (C) A frozen expression map and training-only scaling precede prediction and energy-distance comparison with an unchanged source distribution. Time coordinates, donor count, training seeds, integration resolution and evaluation sampling provide complementary sensitivity checks. The arrays describe samples, not tracked cells; neither task observes diabetic-foot-ulcer healing or time to closure.

of +0.948. In the geometry audit, the median same-donor temporal distance was 1.967 and the median same-time donor distance was 0.233, giving a time-to-donor ratio of 8.46 (bootstrap 95% interval 6.02–13.11). The mean displacement cosine was 0.948, with a descriptive bootstrap interval of 0.943–0.954. These results show that the current representation and protocol contain a transferable signal for one interval in this acute human cohort, while also showing that the geometry summary is based on only three donors (Tables S2 and S9; Figure 2).

The result is deliberately bounded. The donors are healthy volunteers, the wounds are acute experimental wounds and the target is one observed time point. The analysis does not contain a DFU patient trajectory or a time-to-closure endpoint. “Transfer” here means transfer across the three specified acute-wound donors under a held-out protocol; it does not mean clinical generalization.

A fixed-seed refit assessed numerical resolution and evaluation sampling. Across 25, 50, 100 and 200 RK4 steps, the target energy distance remained 0.705395 and the improvement remained 49.863%; the largest discrepancy from the 200-step coordinates was below 6×10^{-7} in root-mean-square units (Table S13). Repeated evaluation at 250, 500 and 1,000 cells per distribution gave median improvements of 48.46%, 48.39% and 49.26%, respectively. Gain exceeded the same-count split-half reference in all 50 draws at each count (Table S14). These results assess one fitted field and the observed cells; the ranges across draws quantify evaluation sensitivity, not additional donor replication or confidence in clinical generalization.

2.2 Time parameterization changes the held-out result

The linear days axis failed at the held-out Wound7 target, with an improvement of –18.2% relative to stay-still. Rank, square-root days and log days met the specified noise-reference criterion, and none of the four tested axes diverged. The result is consistent with a time-coordinate mismatch in which the early wound interval occupies a small fraction of the linear 0–30 day range. It does not establish log days as a universal biological clock; the transform must be prespecified and tested in a design with more independent time series (Table S3; Figure 3).

2.3 Sample-out comparison favours the shared field in this cohort

Under the common sample-out protocol, shared flow matching had mean energy distance 0.395 and passed in 3/3 folds. The non-neural mean-displacement, nearest-donor and optimal-transport baselines also passed in 3/3 folds. The donor-conditioned model passed in 3/3 sample-out folds but had a higher mean energy distance than the shared field in this comparison. In the six

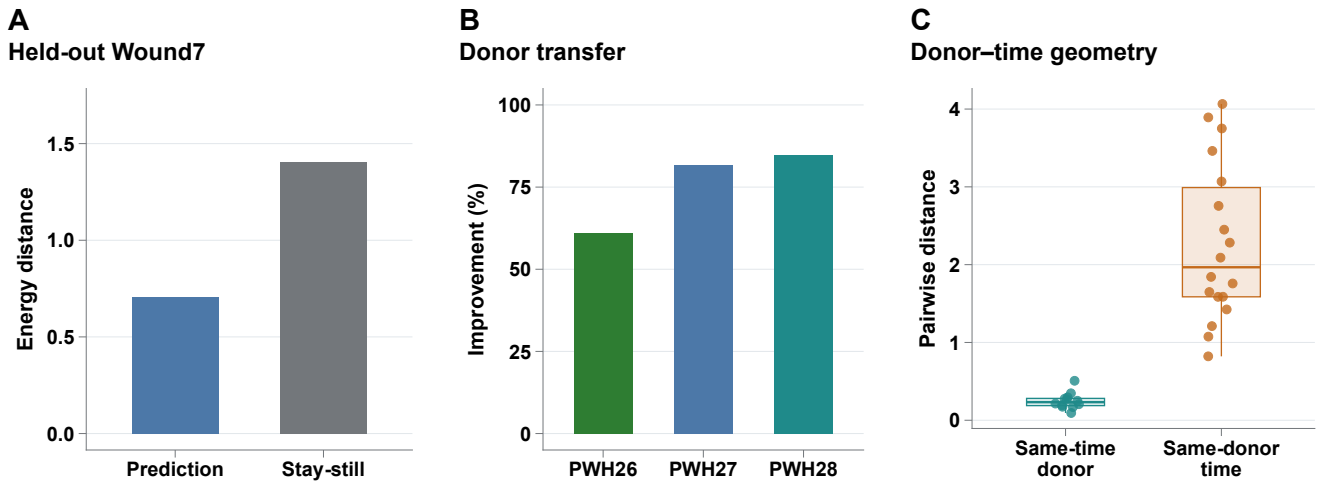


Figure 2. Global Wound7 reconstruction, donor transfer and donor-time geometry. (A) Bars show predicted and stay-still energy distances to pooled Wound7 after excluding that time globally; relative improvement is 100 times (stay-still minus predicted distance) divided by stay-still distance, giving 49.9%. (B) Each bar is a genuinely held-out donor, with all four times retained in the other two donors; the mean relative improvement is 75.9%. Bars are point estimates without confidence intervals. (C) Points show 12 between-donor same-time distances and 18 within-donor between-time distances. Boxes span the interquartile range, centre lines are medians, and whiskers extend to extreme points within 1.5 interquartile ranges. The temporal/donor median ratio is 8.46 (10,000-draw descriptive pair-distance bootstrap 95% interval 6.02–13.11). All distances use training-standardized latent coordinates. Shared donors make these pairs dependent; three healthy donors do not provide DFU prediction validation.

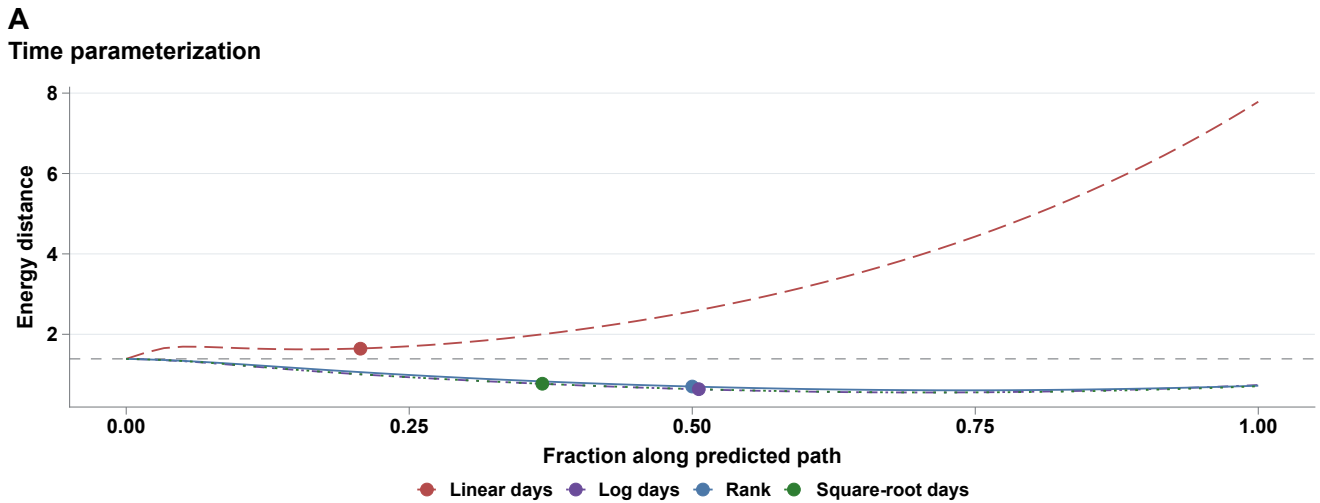


Figure 3. Time-coordinate sensitivity under global Wound7 holdout in three donors. Curves show energy distance along the Wound1-to-Wound30 predicted segment for rank, linear-day, square-root-day and log-day coordinates. The horizontal axis runs from the Wound1 source (0) to the Wound30 coordinate (1); line styles and colours identify the transforms. Dots are placed at each nominal Wound7 fraction with the distance evaluated at the nearest of 61 stored grid positions from 60 RK4 steps. The grey dashed line is the stay-still distance, 1.3879. Linear days gives –18.2% relative improvement at the target; the other axes have gains above the single split-half reference, 0.0061. The full path is a retrospective diagnostic, and its minimum is not substituted for the Wound7 result. No tested axis diverges; curves do not represent confidence bands.

sample-in comparisons, conditioning improved on the shared field in 6/6 cases. This difference between sample-in and sample-out is the central safeguard against interpreting capacity as generalization (Tables S4, S5 and S10; Figure 4).

The data therefore do not justify selecting the conditioned field as the final model class. They support a more limited statement: the conditioned model can fit known donor contexts, while the shared model had the lowest point-estimate sample-out error in the present three-donor benchmark. Neither result identifies the model class that will work in DFU patients.

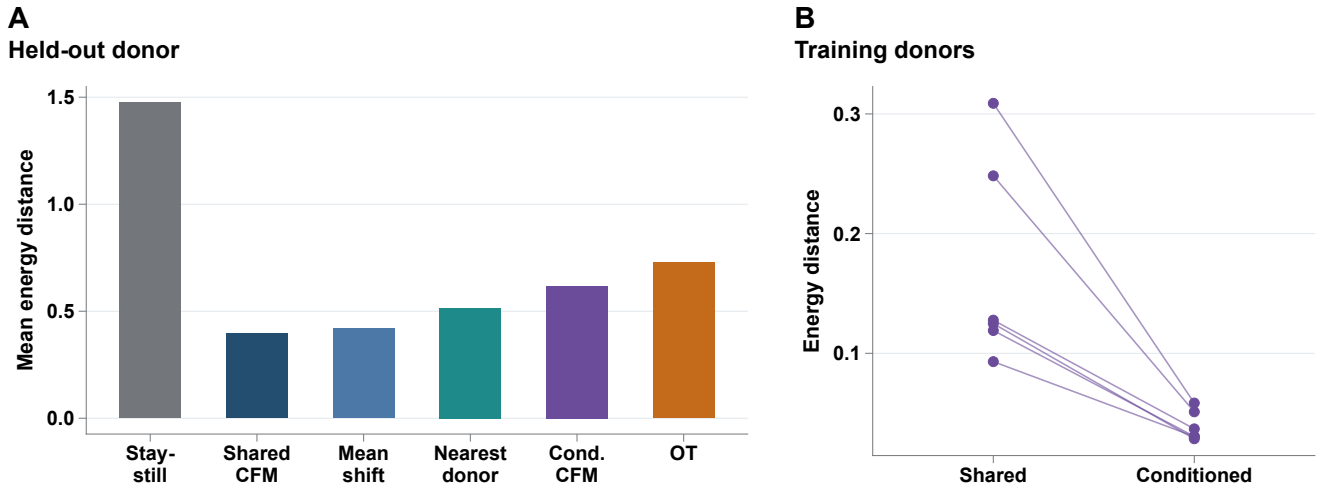


Figure 4. Method comparison across three held-out donors. (A) Bars are arithmetic mean energy distances for stay-still and five predictors evaluated on the same donor folds. CFM means conditional flow matching; OT is the entropic optimal-transport displacement extension. Shared CFM has the lowest point estimate (0.395). Each predictive method improves over stay-still by more than the single split-half target reference in 3/3 folds; this is a descriptive criterion, not a significance test or population method ranking. (B) Each line connects shared and donor-conditioned errors for one of six training-donor comparisons (two retained donors in each of three fits). Conditioning improves all six. These training-donor fits share donors and do not measure transfer to an unseen donor. Neither panel shows uncertainty intervals.

2.4 A second donor improves the two measured points, but $k=1$ already passes

With one training donor, mean held-out energy distance was 0.5496. With two training donors, it was 0.3854. The reduction was positive in each of the three training seeds: 0.164 at seed 0, 0.176 at seed 1 and 0.156 at seed 2. The mean-displacement control also improved at seed 0, from 0.5325 with one donor to 0.4186 with two, so the benefit of a second donor was not unique to the neural field. The mean within-configuration training SD of 0.0717 was 3.13 times the mean split-half reference of 0.0229, showing that the reference omits a material source of run variation (Tables S11 and S12). This makes the aggregate direction reproducible across seeds (Figure 5).

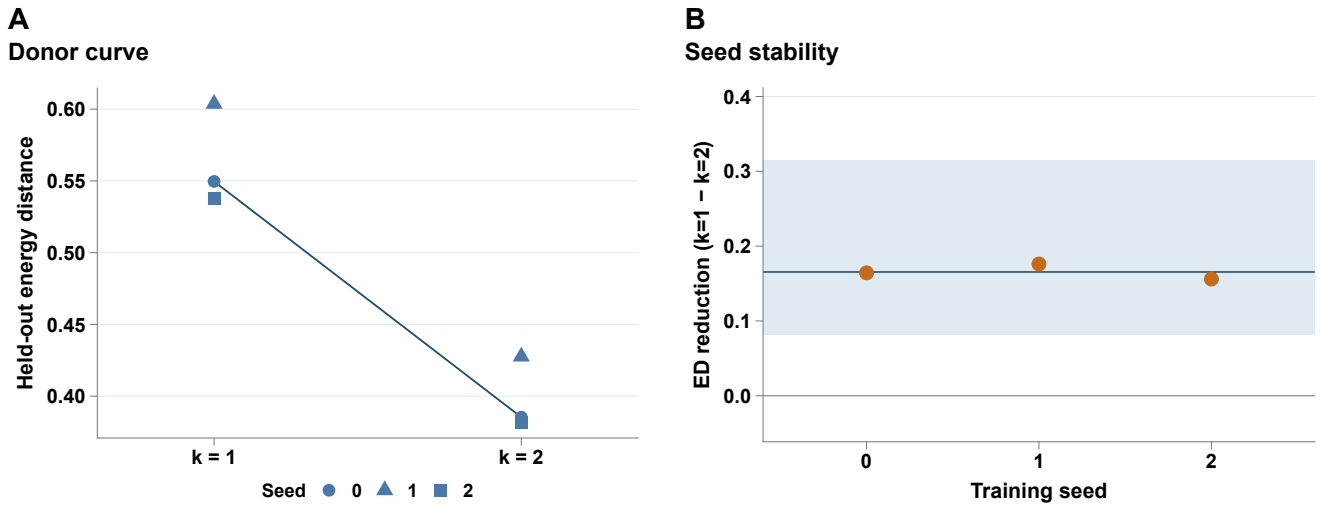


Figure 5. Donor-count comparison and training-seed variation. (A) k is the number of training donors. The line joins seed-0 mean held-out energy distances, 0.5496 at $k=1$ and 0.3854 at $k=2$; symbols identify seeds 0, 1 and 2. At $k=1$, the two single-training-donor choices are averaged within each held donor before equally weighting the three held donors. (B) Points are seed-specific reductions in mean energy distance (ED), $k=1$ minus $k=2$. The horizontal line is the three-seed mean, 0.165. Shading is a percentile 95% interval [0.081, 0.315] from 4,000 resamples of three held-donor blocks after seed averaging; it is not an interval for each seed point. Evaluation subsampling is fixed across seeds. The largest configuration-specific range is 0.368. Overlapping training sets limit interval interpretation, and two k values cannot identify a plateau or required cohort size.

The same analysis shows why the curve cannot be overinterpreted. The paired donor-block reduction averaged 0.165, with a three-block bootstrap 95% interval of [0.081, 0.315]; all three held-out donor blocks improved (Table S6). The largest range for a single donor and training-subset configuration across training seeds was 0.368, while the mean within-configuration standard deviation was 0.0717. The run-to-run variation is therefore large relative to some individual improvements. More importantly, $k=1$ already met the specified energy-distance criterion in all six $k=1$ comparisons. The number of training donors is not an all-or-none success criterion in this dataset, and the two points cannot determine a plateau or a required cohort size. They only establish that adding a second independent context improved the current aggregate error in this pilot.

2.5 Three mouse arms expose model-arm heterogeneity

The all-arm audit gave different results under the same protocol. NDB changed by -1.5% relative to stay-still and did not beat the baseline. PDB improved by $+32.1\%$, but its POD7 target failed an endpoint-distance ordering check, warning against a simple interpolation interpretation without proving an off-manifold trajectory. GDB improved by $+50.6\%$ with a POD7 target satisfying that ordering check. In aggregate, 2/3 arms beat stay-still, 2/3 met the noise criterion and 2/3 satisfied the ordering check. Each arm had one animal per time point, and NDB covered 59.7% of the frozen panel. These arm-level computations are useful for protocol auditing, but they do not establish a mouse biological trajectory, a cross-species equivalence or DFU transfer (Tables S7 and S8; Figure 6).

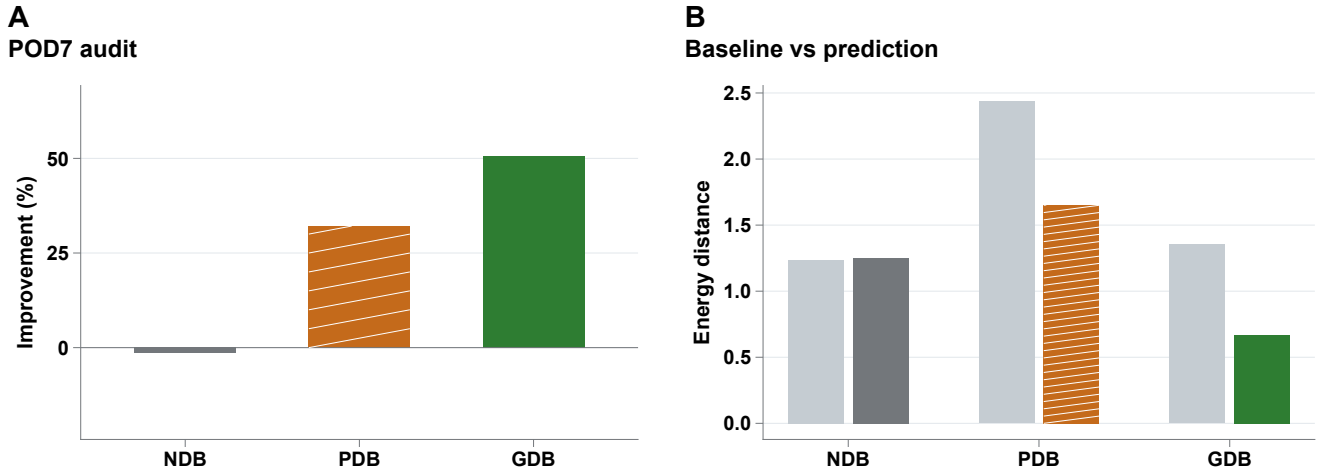


Figure 6. Mouse POD7 computational audit for the source-labelled NDB, PDB and GDB arms. POD7 is excluded from fitting and scaling; rank-time fields trained at POD0, POD2 and POD30 predict from the observed POD2 source. (A) Relative energy-distance improvements over stay-still are -1.5% , $+32.1\%$ and $+50.6\%$. Hatching marks PDB, whose POD7 target fails the endpoint-distance ordering diagnostic. (B) Within each arm, the left grey bar is stay-still and the right coloured bar is prediction; hatching carries the same diagnostic meaning. The screen compares endpoint distances and does not prove or disprove geodesic interpolation. Each arm has one animal per time point and 4,182/7,002 mapped panel genes. Bars have no uncertainty intervals; cell counts do not supply independent animal replication or cross-species clinical validation.

3 Discussion

The experiments support reconstruction of one observed acute-wound interval in a small human cohort. A shared population field improved on an unchanged source distribution when the target time was globally withheld and when a donor was withheld under a separate transfer protocol. These tasks answer different questions: the former lacks target-time observations in every donor, while the latter can learn that time from other donors. Their agreement is informative about this representation and cohort, but does not establish population-wide clinical generalization.

The time-coordinate comparison shows that an observation schedule is part of the model specification. A linear-day axis performed poorly at the specified target even when alternative coordinates improved. This sensitivity is consistent with compressing early changes into a short interval of the global time coordinate. It does not identify a universal biological clock. Distributional transport and flow matching offer flexible parameterizations [10, 16, 20], while biological ancestry remains underdetermined without additional information [8, 22].

The model comparison separates fitting capacity from transfer. Donor conditioning improved the fit to donors represented in training, but the shared field had the lowest mean error on held-out donors. Simple non-neural displacement methods also improved prediction, including when a second training donor was added. The evidence therefore supports useful shared structure without attributing the whole improvement to neural complexity. A future comparison should assess performance in additional donors under a fixed protocol, including simple predictors and methods that model context explicitly [3].

The donor-count experiment identifies a gain at the two measured training sizes. Its aggregate direction persisted across seeds, while individual donor and training-subset configurations varied substantially. One training donor already met the stated empirical criterion. Two observed sizes cannot establish a learning-curve plateau or a required cohort size, and overlapping training sets limit the independence of the fold summaries. Extending the curve requires additional independent donors and a plateau criterion defined before inspecting the expanded data.

The added computational sensitivities distinguish integration error from evaluation variability. Increasing integration resolution changed the target error negligibly for the tested fixed field. Repeated evaluation subsamples at three cell counts also preserved the improvement over the unchanged-source baseline. These findings reduce concern about those particular numerical choices. They do not include uncertainty in the frozen representation, variation between future donors, or a complete repetition of model training. Energy-distance comparison remains conditional on coordinates, cell sampling and the selected target [18].

The human cohort contains only three healthy donors and four times. Cells and repeated subsamples do not increase the number of biological replications [23]. The mouse arms each contain one animal per time point, differ in predictive performance, and have incomplete gene-panel coverage. Their comparison assesses computational behavior across distinct experimental contexts; it does not establish cross-species equivalence. The available data also contain no diabetic-foot-ulcer trajectory with an authoritative time-to-healing endpoint. The next biological test should therefore expand independent-donor evaluation before using an explicitly longitudinal diabetic-foot-ulcer cohort to assess patient-level staging or prognosis.

4 Methods

4.1 Data, representation and join integrity

GSE241132 contains three healthy acute-wound donors, labelled PWH26, PWH27 and PWH28, with Skin, Wound1, Wound7 and Wound30 samples [11, 13]. Among 65,521 loaded cells, 58,823 joined the author metadata, and 12,259 author-annotated fibroblasts entered the temporal analysis. The fibroblast counts were 2,152, 1,012, 3,821 and 5,274 at the four times, respectively. Annotation is joined using sample identity and original cell-row identity before quality filtering; a filtered positional row number is not a valid substitute. Earlier exploratory outputs based on positional mismatching are invalid and contribute no results here.

The expression encoder is frozen from GSE165816 [19]; the human wound cohort was not used to refit it. It maps panel-normalized counts into a $q = 15$ dimensional Gaussian-encoder mean μ . Only 5,383 of 7,002 panel genes are covered in the human wound input (76.9%); absent panel genes are zero-filled. The flow acts on $\mu \in \mathbb{R}^{15}$, not on simplex topic weights $\theta = \text{softmax}(\mu)$. Successful reconstruction is therefore conditional on this incomplete transferred panel and this coordinate system.

For training-cell set $\mathcal{T}$, scaling parameters are fitted coordinatewise on $\mathcal{T}$ only:

$$a_j = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \mu_{ij}, \quad b_j = \left[|\mathcal{T}|^{-1} \sum_{i \in \mathcal{T}} (\mu_{ij} - a_j)^2 \right]^{1/2}, \quad z_{ij} = \frac{\mu_{ij} - a_j}{\tilde{b}_j}. \quad (1)$$

In Equation (1), $\tilde{b}_j = b_j$ for $b_j > 0$ and $\tilde{b}_j = 1$ if $b_j = 0$. The identical transformation is applied to source and target evaluation cells. The target is excluded from estimating a_j, b_j , and all cells of the held-out donor are excluded in donor transfer. Distances from different training folds are consequently expressed in their respective fold-standardized spaces.

4.2 Two distinct held-out tasks

Timepoint reconstruction. All Wound7 cells are withheld from fitting and scaling. Training segments are Skin→Wound1 and Wound1→Wound30 within each donor. The fitted field advances the observed Wound1 source to the specified Wound7 coordinate. The pooled analysis uses three donors. Three additional runs omit one donor and evaluate Wound7 in the remaining donors; these are sensitivity analyses, not predictions for the omitted donor.

Donor transfer. A donor is excluded entirely from fitting and scaling; all four time points in the other two donors are retained. Adjacent within-donor segments Skin→Wound1, Wound1→Wound7 and Wound7→Wound30 are used. The held donor’s Wound1 cells initialize prediction, while their Wound7 cells are used for evaluation only. Because Wound7 is observed in training donors, this task does not test reconstruction of a globally unobserved time point. The method benchmark and donor-count experiment use this latter design. Table S1 lists the exclusions and settings.

4.3 Time coordinates and conditional flow matching

Let $P_t^{(d)}$ denote the empirical distribution of standardized fibroblast coordinates for donor d at day t . Write $s = g(t)$ for transformed time. Rank time assigns $(0, 1/3, 2/3, 1)$ to days $(0, 1, 7, 30)$, and the other tested transforms are

$$g_{\text{days}}(t) = t/30, \quad g_{\text{sqr}}(t) = \sqrt{t/30}, \quad g_{\text{log}}(t) = \log(1+t)/\log 31. \quad (2)$$

Equation (2) defines the three day-based alternatives to rank time. For one training donor and segment (s_a, s_b) , endpoints are sampled *independently* from the source and target cells of that donor. Thus the training coupling is the empirical product $\pi_{d,ab} = P_{t_a}^{(d)} \otimes P_{t_b}^{(d)}$. It is not an optimal-transport coupling and does not pair the same biological cell across times. With $\lambda \sim U(0, 1)$,

$$z_\lambda = (1-\lambda)z_a + \lambda z_b, \quad s_\lambda = (1-\lambda)s_a + \lambda s_b, \quad u_{ab} = \frac{z_b - z_a}{s_b - s_a}. \quad (3)$$

Equation (3) defines the interpolated state, time and target velocity. The implemented loss cycles over donor–segment pairs and averages squared error over cells and coordinates:

$$\mathcal{L}_{\text{CFM}}(\phi) = \mathbb{E}_{(d,a,b), \pi_{d,ab}, \lambda} \left[\frac{1}{q} \|v_\phi(z_\lambda, s_\lambda) - u_{ab}\|_2^2 \right]. \quad (4)$$

In Equation (4), the division of the target velocity by $s_b - s_a$ is necessary: the target is velocity per unit transformed time. In particular, compressing the Skin-to-Wound1 segment on the linear-day axis increases its numerical velocity scale. Equations (3)–(4) implement conditional flow matching [9, 10, 21]. They do not solve the Benamou–Brenier minimum-kinetic-energy problem [2] or establish globally optimal transport.

Following the neural ordinary-differential-equation formulation [4], predictions solve an initial-value problem and push forward the empirical source measure:

$$\frac{dz}{ds} = v_\phi(z, s), \quad z(s_a) = z_a, \quad \widehat{P}_{s_b} = (\Phi_{s_a \rightarrow s_b}^\phi)_\# \widehat{P}_{s_a}. \quad (5)$$

Equation (5) defines prediction from the observed source distribution. The network uses two 128-unit SiLU hidden layers and a 32-dimensional sinusoidal time embedding. Training uses Adam with learning rate 10^{-3} , 8,000 steps and batches of 256. Direct target predictions use 50 fixed fourth-order Runge–Kutta steps. The time-axis diagnostic instead uses 60 steps over the full Wound1-to-Wound30 segment and reads the grid point nearest the transformed Wound7 coordinate; it does not select the grid point with the lowest target distance. A separate fixed-field resolution comparison is described below; finite-step agreement does not prove convergence to an exact solution.

4.4 What distributional success does and does not identify

At the ideal population minimizer of the squared loss, the field is a conditional expectation of path velocities, $v^*(z, s) = \mathbb{E}[u_{ab} \mid z_\lambda = z, s_\lambda = s]$, where defined. Under suitable integrability and regularity, the induced marginal path satisfies a continuity equation,

$$\partial_s p_s + \nabla \cdot (p_s v^*) = 0. \quad (6)$$

Equation (6) describes conservation of probability mass in the model. It neither tracks proliferation/death rates nor determines individual-cell ancestry. Multiple couplings and fields can interpolate the same marginals; finite neural optimization adds approximation error. Matching the observed target distribution therefore does not identify a unique biological vector field [8]. With only four measurement times, the values of g between observations are analysis choices rather than an identified biological clock.

4.5 Distance estimand, finite-sample statistic and thresholds

For independent $X, X' \sim P$ and $Y, Y' \sim Q$ with finite first moments, define energy distance without the optional square root [18]:

$$D_E(P, Q) = 2\mathbb{E}\|X - Y\|_2 - \mathbb{E}\|X - X'\|_2 - \mathbb{E}\|Y - Y'\|_2. \quad (7)$$

In Euclidean space, Equation (7) is nonnegative and zero exactly when $P = Q$. The implementation uses the V-statistic, including diagonal zero distances:

$$\begin{aligned} \widehat{D}_E(P_m, Q_n) &= \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n \|x_i - y_j\|_2 - \frac{1}{m^2} \sum_{i=1}^m \sum_{i'=1}^m \|x_i - x_{i'}\|_2 \\ &\quad - \frac{1}{n^2} \sum_{j=1}^n \sum_{j'=1}^n \|y_j - y_{j'}\|_2. \end{aligned} \quad (8)$$

Equation (8) is the exact distance between the two empirical measures, not an unbiased U-statistic for the population distance. For baseline $B = \widehat{D}_E(P_{\text{source}}, P_{\text{target}}) > 0$ and predicted distance D , gain and relative improvement are

$$G = B - D, \quad I = G/B = 1 - D/B. \quad (9)$$

Equation (9) defines the absolute gain and the fraction reported as a percentage. Stay-still means $P_{\text{prediction}} = P_{\text{source}}$. Seeded subsampling caps the quadratic distance calculation (1,500 cells per distribution in the initial temporal and donor-transfer runs; the common benchmark uses its recorded 1,200-cell cap). Counts, caps and per-fold values accompany each comparison; independently run analyses need not have identical stochastic point estimates.

The target is split once into seeded disjoint halves, defining $N = \widehat{D}_E(P_{\text{target},1}, P_{\text{target},2})$. The initial timepoint report asks only whether $G > 0$, while the time-axis, donor-transfer, method and donor-count comparisons use $G > N$. The latter is a sampling-scale heuristic, not a significance test with known type-I error, a repeated-resampling interval, or an error bound. Selecting the best position on a path using target distances is retrospective; the preassigned Wound7 coordinate supplies the held-out result, and the full scan is a diagnostic rather than a tuned clinical predictor.

4.6 Donor context and the non-neural predictors

For a source sample, context concatenates the coordinatewise mean and standard deviation, $c_d = (\bar{z}_d, \text{sd}(z_d)) \in \mathbb{R}^{2q}$. The conditioned field $v_\phi(z, s, c_d)$ uses this context from the training segment's source. At test time, the held donor's source alone supplies c_d ; its target is not used. Context is zeroed in 20% of training batches. Both neural fields use the same donor-segment schedule and endpoint-sampling rule, although successive pseudorandom draws need not be identical. Six training-donor evaluations are reported separately from the three held-out-donor folds.

Let $\delta_d = \bar{z}_{d,7} - \bar{z}_{d,1}$ for a training donor. Mean displacement predicts $z + \bar{\delta}$; nearest-donor displacement predicts $z + \delta_{d^*}$, where d^* maximizes cosine similarity between the source contexts. These are translations, not fitted neural fields. The entropic transport baseline [5] uses pooled training-source and training-target cells, uniform empirical marginals a, b , and normalized squared Euclidean cost C :

$$\Gamma_\varepsilon = \arg \min_{\Gamma \geq 0, \Gamma \mathbf{1} = a, \Gamma^\top \mathbf{1} = b} \left\{ \langle C, \Gamma \rangle + \varepsilon \sum_{ij} \Gamma_{ij} (\log \Gamma_{ij} - 1) \right\}, \quad \varepsilon = 0.05. \quad (10)$$

Equation (10) defines the entropic transport plan fitted to the training endpoints. Sinkhorn iteration is capped at 2,000 steps, with at most 1,500 cells per endpoint. Row-normalized barycentres define training-source displacements; each held-out source cell receives the average displacement of its 15 nearest training-source neighbours. This is an out-of-sample displacement extension of a transport plan, not transport fitted to the held target.

4.7 Donor-count estimand and dependence-aware summaries

For each held donor d and seed $r \in \{0, 1, 2\}$, let $D_{1,d,r}$ average the errors from the two possible single-training-donor choices, and let $D_{2,d,r}$ be the error with both available training donors. Equal weighting then gives

$$R_k = \frac{1}{3 \cdot 3} \sum_{d=1}^3 \sum_{r=0}^2 D_{k,d,r}, \quad \Delta_d = \frac{1}{3} \sum_{r=0}^2 (D_{1,d,r} - D_{2,d,r}), \quad \bar{\Delta} = \frac{1}{3} \sum_d \Delta_d. \quad (11)$$

Equation (11) defines the seed-averaged estimand and donor-block differences. The plotted main curve uses seed 0, with other seeds shown separately. The 4,000-draw percentile interval resamples the three Δ_d blocks after averaging seeds [6]. Overlapping training donors still induce dependence between held-out errors, so even this interval is a descriptive three-block summary, not a population coverage guarantee. Seeds and alternative training subsets are not additional biological replicates. Configuration-specific ranges quantify stochastic instability. Two values of k cannot identify a plateau, a functional learning-curve model or a required number of donors.

4.8 Geometry and mouse computation

Donor displacement alignment is $\cos(\delta_d, \delta_e) = \delta_d^\top \delta_e / (\|\delta_d\| \|\delta_e\|)$. The time-versus-donor contrast compares 18 within-donor temporal distances with 12 between-donor same-time distances. Pair-distance bootstrap intervals are descriptive because pairs share donors. The geometric screen used for endpoints a, b and held time h is

$$J = \mathbf{1}\{\widehat{D}_E(P_a, P_b) \geq \max[\widehat{D}_E(P_a, P_h), \widehat{D}_E(P_h, P_b)]\}. \quad (12)$$

Equation (12) is a distance-ordering diagnostic. It does not imply equality in a triangle inequality, identify a geodesic, or establish that a target lies on a unique direct path. We report it as such rather than as proof of interpolability.

GSE326622 is evaluated separately for the source-labelled NDB, PDB and GDB mouse arms [14]. POD7 is excluded from fitting and scaling; POD0, POD2 and POD30 provide the training time points, with POD2 supplying the prediction source. Each mouse fit uses rank time, 6,000 optimization steps and batches of 256. The human panel's mouse-symbol mapping covers 4,182 of 7,002 genes (59.7%) in each arm, and each arm has one animal per time point. Animal identity and time are therefore confounded. The three arm computations assess protocol behaviour and representation transfer; cell numbers do not supply independent animals or cross-species clinical validation.

4.9 Numerical-resolution and evaluation-sampling sensitivity

The saved human expression coordinates and annotations supply a separate seed-0 pooled fit using the stated training exclusions, 8,000 optimization steps and batch size 256. The fitted field is then fixed. Predictions integrate the same 1,012 day-1 cells with 25, 50, 100 and 200 RK4 steps to the prespecified rank-time day-7 coordinate. Distance calculations use one fixed set of 1,500 day-7 cells and double-precision pairwise distances. Root-mean-square coordinate discrepancies are measured relative to the 200-step result (Table S13).

Evaluation sensitivity fixes the 50-step prediction and varies only cell subsampling. At each count of 250, 500 and 1,000 cells per distribution, 50 seeded draws without replacement use common source and target indices for predicted and unchanged-source distances. Each draw additionally samples two disjoint target subsets of that size for the reference distance. We report median improvement, the central 95% range across draws and the count exceeding the reference (Table S14). These are conditional Monte Carlo summaries. They neither refit the field per draw nor estimate uncertainty across new donors.

Data and reproducibility

Human acute-wound data and author cell annotations are available as GSE241132 [11, 13]. GSE165816 supplies the independently frozen expression model [19]; its DFU clinical labels are not used to train the human temporal fields. Mouse data are from GSE326622 [14]. This secondary computational study recruited no participants and performed no new animal experiments. Source-study consent and ethical approvals remain those of the original investigators. Sample identities, preprocessing exclusions, numerical settings and all reported comparisons are defined in the methods and supplementary tables. Analysis software, the frozen expression model, numerical reports, figure source data and native manuscript sources are supplied in the accompanying software archive, version 0.1.0, authored by Zeyu Fu. Source-study data are obtained from the stated public repositories. The code repository is <https://github.com/PeterPonyu/single-cell-wound-analysis>.

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Supplementary methods and numerical results

S1 Protocol and estimator specification

Table S1 separates the missing-time and missing-donor questions. Training endpoints are paired by donor, while the individual cells drawn from those endpoints are independent. Holding out a sample in an already trained donor is not equivalent to holding out that donor. The representation is frozen before all temporal comparisons.

Table S1. Training and evaluation information for the six temporal protocols. Human data comprise three donors at Skin, Wound1, Wound7 and Wound30; mouse arms use postoperative days (POD) 0, 2, 7 and 30. Scaling follows the listed training exclusions. Source-only context at test time is permitted for donor-conditioned prediction, while the evaluation target is excluded from fitting in each held-out task. The explicit target column distinguishes prediction for an excluded donor from sensitivity analyses evaluated only in retained donors.

Protocol	Training data	Evaluation target and interpretation
Global Wound7 holdout	Skin, Wound1 and Wound30 from the included donors	Their Wound7 distributions; one globally missing measurement time.
Drop-one-donor sensitivity	Same time exclusions after removing one donor	Wound7 in the retained donors; sensitivity to donor composition, not prediction of the omitted donor.
Donor transfer	All four times in the other two donors	Omitted donor’s Wound7 from their Wound1 source; observed interval in the training cohort.
Method benchmark	Same donor-transfer folds	Shared/conditioned fields and three non-neural predictors; training-donor fits shown separately.
Donor-count curve	One or two of the available training donors	Same held donor and target; seeds and subsets do not add independent donors.
Mouse POD7 audit	POD0, POD2 and POD30 in one source arm	POD7 from POD2; one animal per time point confounds time and individual.

S2 Numerical prediction sequence

The reproducible evaluation order is: (1) resolve cell/sample identities and select the lineage; (2) define the held-out cells; (3) fit standardization only on training cells; (4) sample training endpoints and optimize the specified objective; (5) transport source cells with the fixed numerical integrator; (6) compute target distance, stay-still distance and the single split-half reference; (7) aggregate within the stated donor, seed and subset units. Target-distance path scans are diagnostics after prediction and are not used to select the reported target coordinate.

For completeness, the fixed-step RK4 update for step h is

$$\begin{aligned}
 k_1 &= v(z_n, s_n), & k_2 &= v(z_n + hk_1/2, s_n + h/2), \\
 k_3 &= v(z_n + hk_2/2, s_n + h/2), & k_4 &= v(z_n + hk_3, s_n + h), \\
 z_{n+1} &= z_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4), & h &= (s_b - s_a)/50.
 \end{aligned} \tag{S1}$$

Equation (S1) gives the numerical update for the direct target predictions. The conditioned field holds the source-derived context fixed throughout this integration. Under standard smoothness assumptions RK4 has fourth-order global convergence, but this property does not establish the actual error of the fitted field or certify the chosen step count in these experiments.

S3 Limits of the noise and geometric checks

The finite empirical energy distance is nonzero when two independently sampled subsets represent the same population. The split-half reference therefore indicates one realized scale of target-sampling variation. It changes with cell number, latent scaling and random split, and does not include model-fit uncertainty. Comparing a prediction gain to it is a practical diagnostic with no established frequentist size in this analysis.

Likewise, the endpoint-distance ordering condition cannot show that a point lies on a line or a geodesic. In a metric space, even being closer to both endpoints than the endpoints are to each other is only a necessary-looking geometric screen, not an interpolation identity; here the unsquared-root energy quantity is used by convention and no triangle-inequality argument is invoked. PDB is consequently labelled as failing the ordering screen, not as a proven off-path biological state.

The fitted transport moves probability mass between empirical distributions; it does not model absolute cell expansion, death, migration into a biopsy or clinical closure. These unobserved processes and the non-uniqueness of couplings prevent a mechanistic interpretation of an individual streamline.

S4 Uncertainty and numerical results

Tables S2–S9 provide target errors, time-axis comparisons, the six-entry benchmark including stay-still, donor-specific method errors, the seed-averaged donor curve, mouse-arm contrasts and descriptive geometry intervals. All five predictive methods meeting a criterion in three folds is not five independent replications, because they share test donors. The lowest observed mean is reported without a population-level method ranking.

In the donor curve, two alternative single-donor fits contribute to each $k = 1$ donor/seed value. The three-seed mean improvement is then aggregated over three held donors. Because training sets overlap between folds, the block interval does not estimate calibrated population or predictive uncertainty. Extending the curve requires new independent donors before any plateau rule can be evaluated. There is no statistically identified minimum cohort size from the two observed k values. Table S10 reports all six training-donor comparisons. Tables S11 and S12 separate mean-displacement controls, seed-level means and configuration-specific training variability. Table S8 lists the cell counts behind each mouse computation; each cell group still represents only one animal. The retained-donor sensitivity rows in Table S2 evaluate the two donors left in the fit, not the excluded donor.

Table S2. Human Wound7 evaluation in GSE241132. ED denotes the empirical energy-distance V-statistic in training-standardized latent space; lower values indicate closer predicted and observed distributions. The pooled row excludes Wound7 globally. Drop-one rows exclude the named donor and evaluate pooled Wound7 only in the two retained donors. Transfer rows instead evaluate the excluded donor after fitting all four time points of the other donors. Improvement is 100 times (stay-still ED minus predicted ED) divided by stay-still ED. Split-half ED uses one seeded target split and is not a confidence limit. Every row is a point estimate; these dependent fits do not create additional donors.

Evaluation	Stay-still ED	Predicted ED	Split-half ED	Improvement
Pooled time holdout	1.4069	0.7054	0.0043	49.86%
Drop PWH26; retained pair	1.3357	0.7434	0.0043	44.34%
Drop PWH27; retained pair	1.2780	0.6258	0.0128	51.04%
Drop PWH28; retained pair	1.5460	0.7638	0.0080	50.59%
PWH26 transfer	1.8221	0.7110	0.0171	60.98%
PWH27 transfer	1.5355	0.2786	0.0096	81.85%
PWH28 transfer	1.0704	0.1634	0.0418	84.73%

Table S3. Four time coordinates under global Wound7 holdout. The reference distance is 1.3879 and the split-half distance is 0.0061. ED denotes empirical energy distance in training-standardized latent space. Wound7 time is the global transformed coordinate; the reported target distance is evaluated at the nearest point of the 60-step Wound1-to-Wound30 scan. Improvement is 100 times the gain divided by stay-still distance. Rows use the same three donors, with Wound7 excluded globally; the path minimum is not substituted for target error, and no interval is shown.

Time coordinate	Wound7 time	Target ED	Improvement	Diverged
rank	0.6667	0.6994	49.61%	False
days	0.2333	1.6411	-18.24%	False
sqrt days	0.4830	0.7695	44.56%	False
log days	0.6055	0.6344	54.29%	False

Table S4. Common Wound1-to-Wound7 benchmark across the three held-out human donors. ED is empirical energy distance in each training-standardized fold; reported means equally weight donors. Gain is stay-still ED minus method ED. Pass count records gains above the single split-half target reference, a descriptive criterion rather than a significance test. CFM denotes conditional flow matching and OT the entropic optimal-transport displacement extension. All methods share test donors; point estimates without confidence intervals do not establish a population ranking.

Method	Mean ED	Mean gain	Pass count
Stay-still	1.4760	0.0000	0/3
Shared CFM	0.3947	1.0813	3/3
Mean displacement	0.4186	1.0574	3/3
Nearest donor	0.5147	0.9613	3/3
Conditioned CFM	0.6176	0.8584	3/3
Entropic OT extension	0.7284	0.7476	3/3

Table S5. Empirical energy distances for each method and held-out human donor in the common benchmark. Each column uses the other two donors for fitting and scaling, the named donor's Wound1 cells as source and its Wound7 cells only for evaluation. Lower is better; stay-still retains the source distribution. CFM means conditional flow matching; OT is the entropic transport displacement extension. Entries are point estimates with a 1,200-cell distance cap. Training draws and this cap differ from separately run transfer experiments, so those point estimates need not coincide.

Method	PWH26	PWH27	PWH28
Stay-still	1.8221	1.5355	1.0704
Shared CFM	0.7110	0.2881	0.1850
Mean displacement	0.5674	0.3496	0.3388
Nearest donor	0.6792	0.4318	0.4331
Conditioned CFM	1.4730	0.1990	0.1809
Entropic OT extension	0.7201	0.6228	0.8423

Table S6. Seed-averaged donor-count comparison for shared flow matching. ED is empirical energy distance and k is the number of training donors. At k=1, the two single-training-donor choices are averaged within each held donor and seed; each row then averages seeds 0, 1 and 2. Reduction is k=1 ED minus k=2 ED. The equally weighted mean reduction is 0.1655, with a 4,000-draw percentile 95% interval [0.0813, 0.3146] from three held-donor blocks. Training sets overlap; this descriptive interval is not a population coverage guarantee or evidence of a learning-curve plateau.

Held donor	k=1 ED	k=2 ED	Reduction
PWH26	1.0732	0.7586	0.3146
PWH27	0.3373	0.2368	0.1005
PWH28	0.2807	0.1994	0.0813

Table S7. Mouse POD7 audit for the source-labelled NDB, PDB and GDB arms. Each field uses rank time and POD0, POD2 and POD30 for fitting/scaling, then predicts from POD2 to excluded POD7. ED is empirical energy distance; improvement is 100 times (stay-still ED minus predicted ED) divided by stay-still ED. Each arm covers 4,182/7,002 human-panel genes and has one animal per time point. The ordering check asks whether endpoint distance is at least both endpoint-to-target distances; it is not proof of a geodesic. Values have no confidence intervals, and cell counts do not provide animal replication.

Arm	Stay-still ED	Predicted ED	Improvement	Ordering check
NDB	1.2341	1.2525	-1.49%	Yes
PDB	2.4345	1.6526	32.12%	No
GDB	1.3532	0.6682	50.62%	Yes

Table S8. Fibroblast cell counts and target-sampling reference for each mouse arm. POD means postoperative day. POD0, POD2 and POD30 enter training; POD7 is the held-out target. Every arm/timepoint cell group comes from one animal, so these counts quantify distribution sampling and not biological replication. The final column is one seeded split-half POD7 energy distance in the training-standardized space. NDB, PDB and GDB are retained source-study arm labels. All arms share 4,182/7,002 mapped panel genes.

Arm	POD0	POD2	POD7 held out	POD30	Split-half ED
NDB	2818	8177	12914	550	0.00544
PDB	1197	471	2500	813	0.01091
GDB	223	2400	11590	1402	0.00617

Table S9. Descriptive geometry for three healthy acute-wound donors. The first estimate divides the median of 18 within-donor between-time energy distances by the median of 12 between-donor same-time distances. The second is the mean of three pairwise cosines between donor Wound1-to-Wound7 mean-displacement vectors. Endpoints are percentile 95% limits from 10,000 resamples of pairwise distances within class or of cosine pairs. Pairs share donors, so these intervals do not provide independent donor replication or a general population ordering.

Summary	Estimate	Lower endpoint	Upper endpoint
time to donor median ratio	8.4570	6.0173	13.1114
mean displacement cosine	0.9477	0.9432	0.9536

Table S10. All six training-donor comparisons for shared and donor-conditioned flow matching. The first column identifies the donor excluded from that fit; the second is a retained training donor whose Wound1-to-Wound7 prediction is evaluated. ED is empirical energy distance in the fold-standardized coordinates. Both source and target of the evaluated donor were available during fitting, so these are fitting-capacity comparisons rather than donor-transfer tests. Conditioned errors are lower in all six rows. The repeated use of three donors prevents interpreting the rows as six independent biological replicates.

Excluded donor	Evaluated training donor	Stay-still ED	Shared ED	Conditioned ED
PWH26	PWH27	1.6257	0.1277	0.0369
PWH26	PWH28	1.0837	0.0930	0.0305
PWH27	PWH26	1.7375	0.2483	0.0511
PWH27	PWH28	1.0227	0.1248	0.0286
PWH28	PWH26	1.7873	0.3089	0.0584
PWH28	PWH27	1.6191	0.1190	0.0306

Table S11. Donor-count means by training seed and non-neural control. CFM denotes conditional flow matching, ED empirical energy distance and k the number of training donors. At k=1, two available single-donor fits are averaged within held donor; at k=2 one fit uses both available donors. Means then equally weight the three held donors. Reduction is k=1 minus k=2. CFM evaluation subsamples are fixed across seeds 0, 1 and 2; the mean-displacement control is recorded at seed 0. All nine CFM runs per seed and all six/three control runs at k=1/k=2 meet the descriptive split-half criterion. The control also improves with a second donor.

Predictor	Seed	k=1 mean ED	k=2 mean ED	Reduction
Shared CFM	0	0.5496	0.3854	0.1643
Shared CFM	1	0.6037	0.4276	0.1761
Shared CFM	2	0.5380	0.3819	0.1560
Mean displacement	0	0.5325	0.4186	0.1139

Table S12. Training stochasticity for every held-donor/training-subset configuration. Mean ED, sample SD and range summarize the same fit repeated at seeds 0, 1 and 2; range is maximum minus minimum. Evaluation subsampling and the target split are fixed, so variation reflects training randomness. Averaging the nine configuration SDs gives 0.0717, 3.13 times the mean split-half reference 0.0229; the maximum range is 0.3683. Seeds and overlapping subsets are not independent donors and the single split-half criterion omits this training variability.

Held donor	Training donors	k	Mean ED	SD	Range
PWH26	PWH27	1	1.3278	0.2070	0.3683
PWH26	PWH28	1	0.8186	0.0066	0.0125
PWH26	PWH27, PWH28	2	0.7586	0.1238	0.2334
PWH27	PWH26	1	0.2038	0.0291	0.0581
PWH27	PWH28	1	0.4709	0.1043	0.1896
PWH27	PWH26, PWH28	2	0.2368	0.0516	0.1021
PWH28	PWH27	1	0.3322	0.0878	0.1757
PWH28	PWH26	1	0.2293	0.0325	0.0605
PWH28	PWH26, PWH27	2	0.1994	0.0029	0.0057

Table S13. Fixed-field Runge–Kutta resolution comparison for pooled day-7 holdout. A single seed-0 field was refitted with the specified 8,000 optimization steps and 256-cell batches on the saved expression coordinates, excluding day 7 from fitting and scaling. All rows use the same 1,012 day-1 cells and the same 1,500 sampled target cells. Energy distance is the empirical V-statistic evaluated in double precision; improvement is relative to the same unchanged-source baseline (1.406924). RMS is the root-mean-square coordinate difference from the 200-step solution; it is a numerical discrepancy, not a confidence interval. The comparison is conditional on one field and finite-precision arithmetic and does not establish formal convergence to an exact ODE solution.

RK4 steps	Energy distance	Improvement	RMS vs 200 steps
25	0.705395409	49.862585%	5.46e-07
50	0.705395449	49.862582%	4.76e-07
100	0.705395463	49.862581%	5.36e-07
200	0.705395452	49.862582%	0

Table S14. Evaluation-sampling sensitivity for the same fixed field with 50 RK4 steps. For each cell count, 50 seeded draws sample source and target cells without replacement; prediction and unchanged-source comparisons use identical source indices and identical target indices within a draw. An additional pair of disjoint target subsets at that count defines a split-half reference. Percentages summarize relative improvement over unchanged source; the central 95% draw range measures Monte Carlo sensitivity conditional on this cohort and field, not patient-level confidence. The final column counts draws whose gain exceeds the split-half reference. Cell count and resampling do not increase the three-donor biological sample size.

Cells per distribution	Draws	Median improvement	Central 95% draw range	Gain exceeds reference
250	50	48.46%	44.24 to 51.57%	50/50
500	50	48.39%	46.22 to 51.69%	50/50
1000	50	49.26%	47.86 to 50.60%	50/50